MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, MODULAR, TYPE IV, CONNECTOR, 150 CONTACT, OFFSET

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-28754.

AMSC N/A FSC 5935
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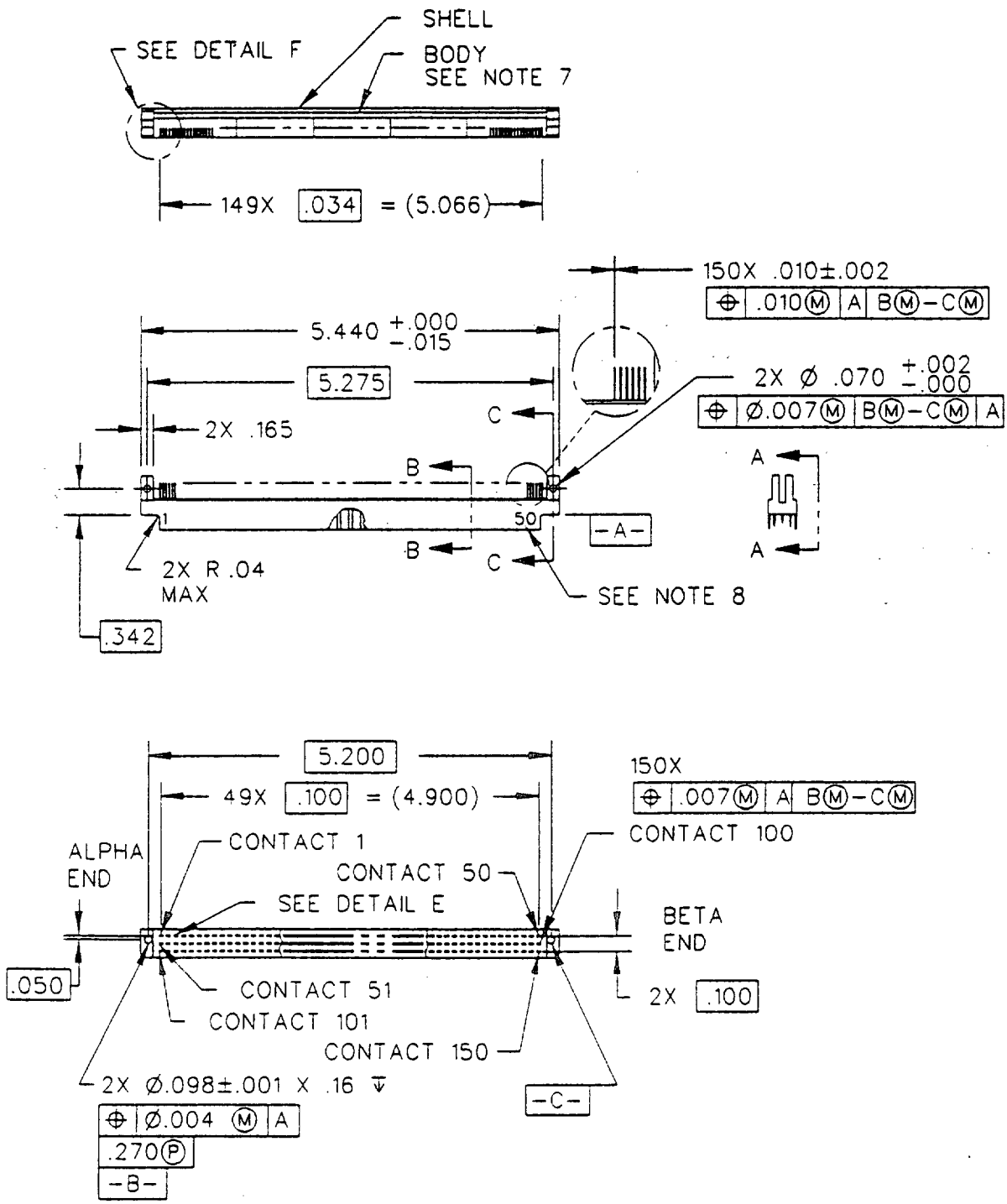
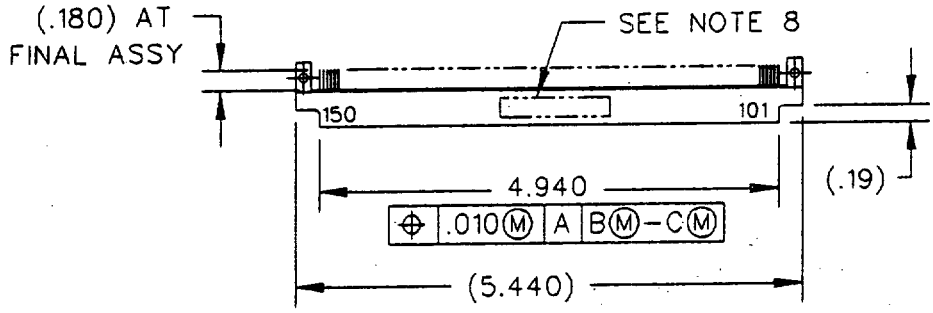
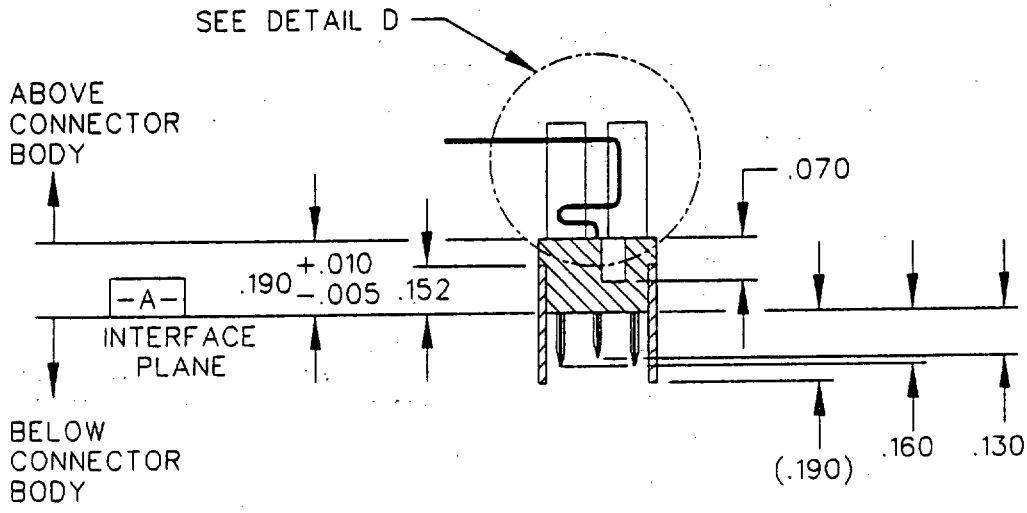


FIGURE 1. Dimensions and configurations.

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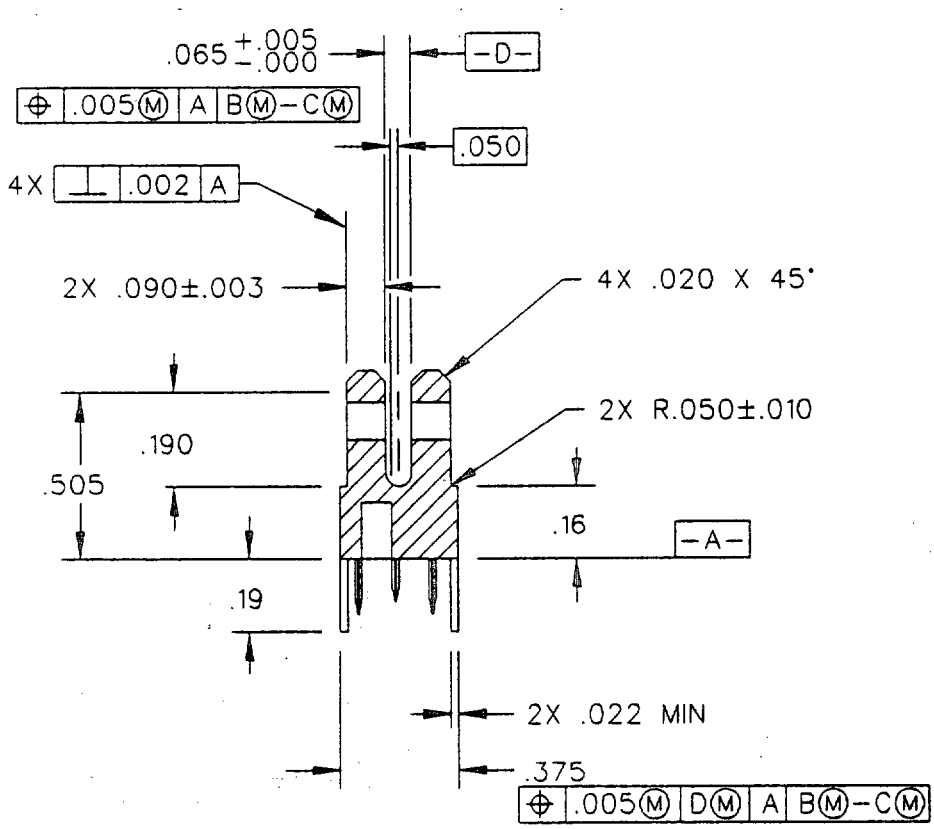
VIEW A–A



SECTION B–B

FIGURE 1. Dimensions and configurations - Continued.

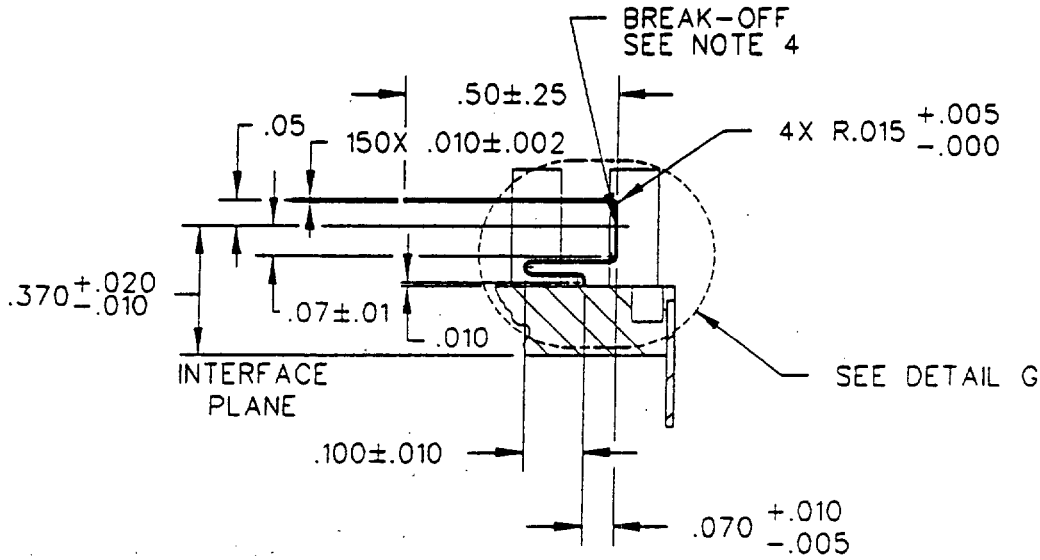
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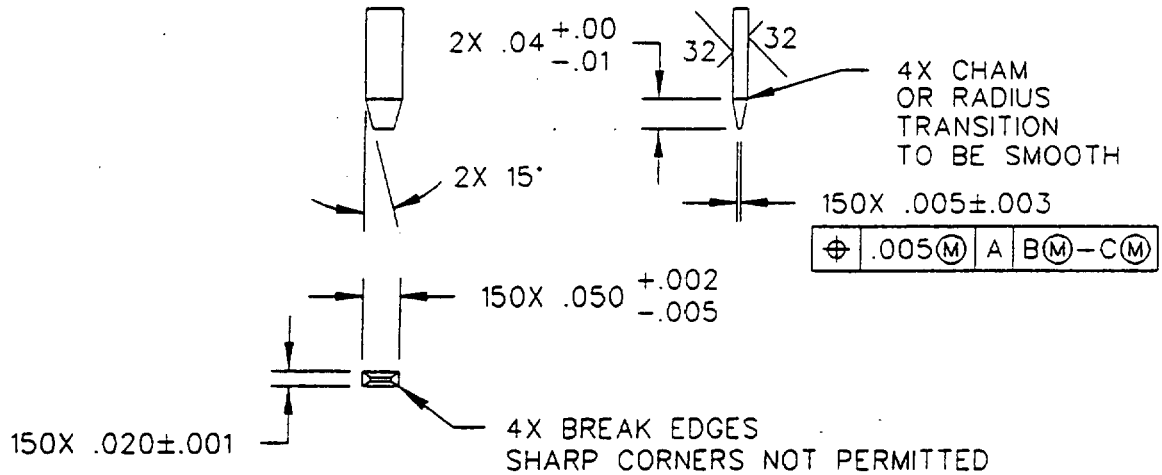
SECTION C-C

FIGURE 1. Dimensions and configurations - Continued.

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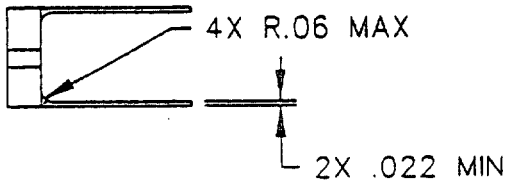
DETAIL D



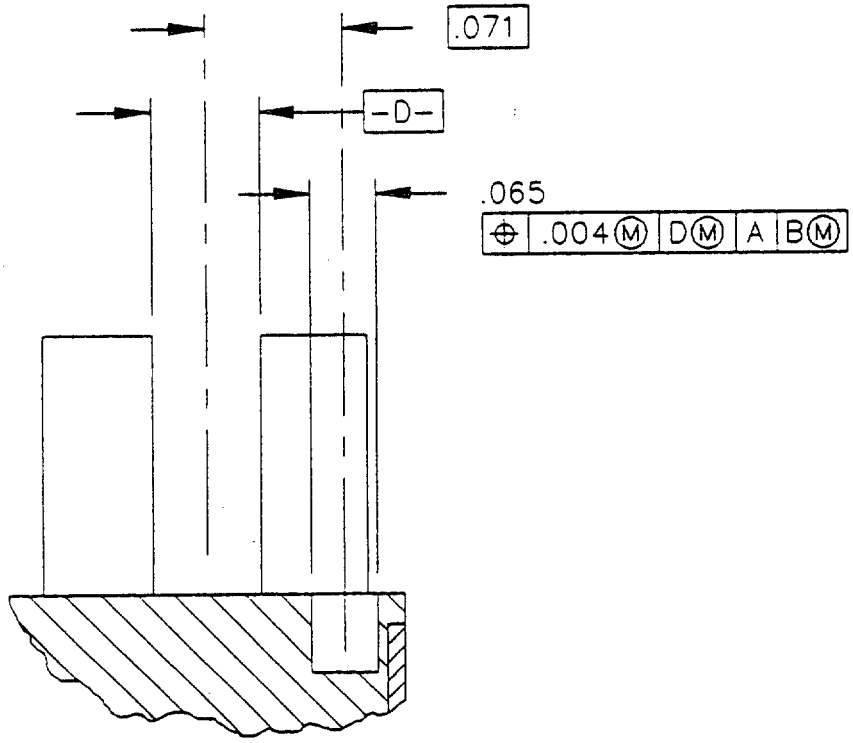
DETAIL E

FIGURE 1. Dimensions and configurations - Continued.

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DETAIL F
SHELL SHOWN BODY
OMITTED FOR CLARITY



DETAIL G
CONTACT TAIL OMITTED FOR CLARITY

FIGURE 1. Dimensions and configurations - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
0.001	0.025	0.050	1.270	0.152	3.861	0.505	12.827
0.002	0.051	0.060	1.524	0.160	4.064	4.940	125.476
0.004	0.102	0.065	1.651	0.165	4.191	5.200	132.080
0.005	0.127	0.070	1.778	0.180	4.572	5.275	133.985
0.007	0.178	0.083	2.108	0.190	4.826	5.440	138.176
0.010	0.254	0.090	2.286	0.250	6.350		
0.020	0.508	0.098	2.489	0.342	8.687		
0.022	0.559	0.100	2.540	0.370	9.398		
0.034	0.864	0.130	3.302	0.375	9.525		
0.040	1.016	0.148	3.759	0.500	12.700		

NOTES:

1. Dimensions are in inches.

2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.40 mm.

3. Unless otherwise specified, tolerances are plus or minus 0.005 (0.127 mm) inch for three place decimals, plus or minus 0.01 (0.25 mm) inch for two place decimals and plus or minus 0° 30′ for angles.

4. The carrier strip shall remain attached to the contact tails to facilitate shipping. The conductors shall be scored at the point indicated to facilitate removal at final assembly, and shall break off in 5 plus or minus 2 flexure cycles of plus or minus 45° from nominal. The carrier strip is not required to be continuous.

5. Except for solder coating dimensional limits apply after plating.

6. Flash around blade contacts not to exceed 0.010 (0.254 mm) inch from the surface of the insulator and shall be firmly attached.

FIGURE 1. Dimensions and configurations - Continued.

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7. The connector body shall be made of 5 identical sections, each containing 30 contacts.

8. Contact shield shall be marked with contact numbers 1, 50, 101, and 150 in contrasting color ink 0.06 (1.52 mm) inch high. Connectors shall be marked in approximate location shown with part number, manufacturer's CAGE number and date code with contrasting ink in characters 0.06 inch (1.52 mm) high. All markings shall be permanently and legibly marked in accordance with MIL-STD-1285.

9. Plating defects on the carrier strip above the break-off line are acceptable, provided that inspection testing of the connector is not effected.

FIGURE 1. Dimensions and configurations - Continued.

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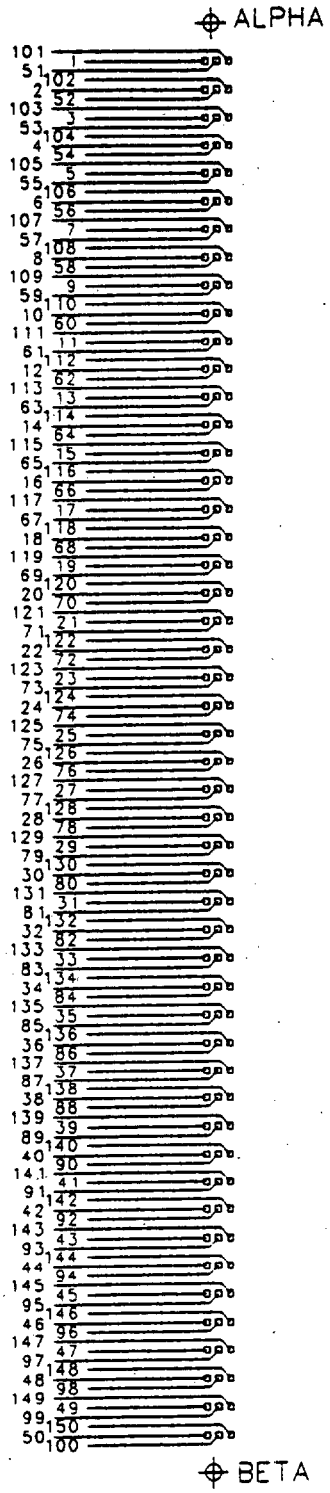


FIGURE 2. Contact pattern numbering.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Materials and finishes: The contacts shall be brass, QQ-B-613, alloy 260, temper 1/2 hard. The contacts shall be finished as follows:

Above the connector body, the contacts shall be finished by the following steps:

1) Steps 1, 2, 4, 5 and 7 - MIL-P-81728 cleanliness.

2) Step 3 - Copper plate, MIL-C-14550, class 4, 0.0001 inch (0.0025 mm) minimum.

3) Step 6 - Tin (90% Sn - 10% Pb) plate, MIL-P-81728, 0.0003 - 0.0005 inch (0.0076 - 0.0127 mm) thickness.

4) Step 8 - Bake, MIL-P-81728, section 3.3.4.

5) Step 9 - Form the contacts.

6) Step 10 - Reflow in hot (235°C) peanut oil, 30 seconds. Quality assurance of MIL-P-81728 shall apply. No indication of dewetting or poor solderability shall be allowed.

7) Step 11 - Rinse in Freon TF or mineral spirits, TT-T-291.

8) Step 12 - Selectively coat the area indicated on the drawing with solder, QQ-S-571, SN60 or SN63, in accordance with MIL-STD-2000 to a thickness of 0.0005 - 0.003 inch (0.0127 - 0.076 mm). Use flux, MIL-F-14256, type RMA, as required.

Alternate finishes using the same or similar materials are allowed on approval of the qualifying activity.

The connector body shall be molded, plastic, diallylphthalate resin, gl-filler, MIL-M-14, type GD1-30F or type SDG-F.

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The connector shell shall be aluminum alloy 6061, QQ-A-200/8, T6, T6510 or T6511.

Aluminum alloy shall be anodic coating, MIL-A-8625, type III, class 2, dyed black. The inside surface of the 0.098 (2.489 mm) diameter hole shall be 90% free of anodize. The anodic coating shall be 0.0020 plus or minus 0.0005 thick on all flat surfaces. Corners shall have a slight radius to facilitate coverage.

Solderability: In accordance with MIL-C-28754.

ELECTRICAL REQUIREMENTS:

Dielectric withstanding voltage: The connector shall meet the requirements of MIL-STD-1344, Method 3001, test condition I for 350 VAC (RMS) 60 Hz, or 500 VDC. The point of measurement shall be between adjacent contacts.

Low level signal contact resistance: Voltage drop shall not exceed 1.2 millivolts. Wire size is not applicable.

Contact resistance: Voltage drop shall not exceed 30 millivolts at 3 Amps current.

Current rating: 3 Amps max per contact.

Voltage rating: 300 Volts, AC (RMS), at sea level.

Durability: In accordance with MIL-C-28754, voltage drop shall not exceed 35 mV after testing.

Military part number: M28754/99.

Qualification information: The connector shall meet the qualification test of MIL-C-28754, table IV, type IV, except for Subgroup IV, Fin/Header torque, Fin cantilever load, and Guide rib strength test.

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Number of units to be inspected: Twelve (12) units of each dash number to be qualified shall be inspected. See MIL-C-28754 Appendix for exceptions when qualifying similar parts, or contact the qualification activity.

Backplane contacts: Low Insertion Force (LIF) contacts in accordance with MIL-C-28859 or MIL-C-28754 are strongly recommended for use with this connector.

Compatibility: This connector may be used with the frame described in MIL-C-28754/96.

Supersession data: Supersedes Naval Air Systems Command Drawing Number SK-MEL-4610, Code Identification Number 30003.

Custodian:
NAVY - SH

User activites:
NAVY - AS

Agent:
NWSC, Crane

Preparing activity:
NAVY - SH

(Project 5935-N350-03)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-28754/99(SH)	Connectors, Electrical, Modular, Type IV, Connector, 150 Contact, Offset

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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